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Advanced Software Engineering (IASE)
Assignment 1

FIFA World Cup 2026 Betting Console Application

Submitted by:

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Declaration of GenAI Use

I, **Rohit Sonejee**, student ID **4756184**, declare that I used generative AI tools (specifically Kimi and Gemini) in the development of this assignment. Below is a transparent breakdown of what was done independently versus with AI assistance.

Part 1: Independently Written Code (by me)

The core application structure was written by me with guidance from the assignment document:

- Data classes (`WorldCupData`, `Group`, `Team`, `Match`, `Knockout`)
- Basic menu loop with options
- Standings calculation logic (points: 3/1/0, goal difference tie-breaker)
- Basic betting system
- JSON parsing using `kotlinx.serialization`
- JUnit tests for core logic

Sample output of my original code:

```
===== FIFA World Cup 2026 - Betting Console =====
1) Show Standings
2) Show Matches
3) Place Bets
4) Show Betting Score
5) Exit
==================================================
Choose an option (1 to 5): 3
Available groups:
1. Group A
2. Group B
...
Which group do you want to bet on? 2
Placing bets for Group B...
```

Listing 1: Original Core Menu Output

Part 2: AI-Assisted Enhancements

After completing the core requirements, I used generative AI tools to enhance and extend the application with the following features:

Feature	**Description**	**AI Role**
Filled JSON with scores & odds	Generated realistic match results and betting odds for all 80 matches	Complete dataset generation
Tournament phase flow	Progression from Group Stage to Final with automatic state transitions	Designed phase logic
Knockout stage menus	Different menus for Round of 32, Round of 16, Quarter-finals, Semi-finals and Final	Implemented UI logic

Money betting system	Bet amounts, odds, payout calculation, win/loss summaries and bankroll tracking	Added betting features
Cumulative records	Team statistics continue updating across all tournament phases	Created record service
3-column group display	Compact side-by-side group layout for cleaner menu presentation	Refactored display system
Smart group finder	Accepts inputs such as Group A, A, a or 1	Added helper parser

Part 3: What I Learned

Through this assignment, I gained deeper understanding of:

- Kotlin data classes, enums, `when` expressions and null safety
- `kotlinx.serialization` for JSON handling
- Gradle build system and dependency management
- Separation of concerns (model, service, UI layers)
- State machine design for managing application phases

Latest Program Execution Output

The following shows the complete execution of the final enhanced application:

```
=======================================================
BEFORE GROUP STAGE
=======================================================
1) Show Standings (Pre-Tournament)
2) Show Matches
3) Place Bets
4) View / Modify Bets
5) Proceed to Group Stage
6) Exit
=======================================================
Choose: 3

Choose by number (1-12) or letter (A-L)

[1/A] Group A              [2/B] Group B              [3/C] Group C
   Czech Republic             Bosnia & Herzegovina       Brazil
   South Korea                Canada                     Haiti
   Mexico                     Qatar                      Morocco
   South Africa               Switzerland                Scotland

[4/D] Group D              [5/E] Group E              [6/F] Group F
   Australia                   C u r a ao                  Japan
   Paraguay                   Ecuador                    Netherlands
   Turkey                     Germany                    Sweden
   USA                        Ivory Coast                Tunisia

[7/G] Group G              [8/H] Group H              [9/I] Group I
   Belgium                    Cape Verde                 France
   Egypt                      Saudi Arabia               Iraq
   Iran                       Spain                      Norway
   New Zealand                Uruguay                    Senegal

[10/J] Group J             [11/K] Group K             [12/L] Group L
   Algeria                    Colombia                   Croatia
   Argentina                  DR Congo                   England
   Austria                    Portugal                   Ghana
   Jordan                     Uzbekistan                 Panama

Group? (0 = back): j

--- Group J Matches ---
0. Back to group selection
1. Argentina vs Algeria (2026-06-16) [No bet]
    Venue: Kansas City
    Odds: 1=1.25 | 0=5.5 | 2=12.0
2. Austria vs Jordan (2026-06-16) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=1.35 | 0=4.5 | 2=9.0
3. Argentina vs Austria (2026-06-22) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=1.4 | 0=4.5 | 2=8.0
4. Jordan vs Algeria (2026-06-22) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=3.4 | 0=3.3 | 2=2.15
5. Algeria vs Austria (2026-06-27) [No bet]
    Venue: Kansas City
    Odds: 1=2.6 | 0=3.2 | 2=2.8
6. Jordan vs Argentina (2026-06-27) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=15.0 | 0=6.5 | 2=1.18

Match? (0 = back): 1

Selected: Argentina vs Algeria
Venue: Kansas City
Odds: 1 (Home) 1.25 | 0 (Draw) 5.5 | 2 (Away) 12.0

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
1
```

```
Selected: 1 (Home Win) @ odds 1.25
Enter bet amount ($): 500

Potential win: $625.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $500.0 on 1 (Home Win) @ 1.25

--- Group J Matches ---
0. Back to group selection
1. Argentina vs Algeria (2026-06-16) [Bet: 1 (Home Win) $$500.00]
    Venue: Kansas City
    Odds: 1=1.25 | 0=5.5 | 2=12.0
2. Austria vs Jordan (2026-06-16) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=1.35 | 0=4.5 | 2=9.0
3. Argentina vs Austria (2026-06-22) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=1.4 | 0=4.5 | 2=8.0
4. Jordan vs Algeria (2026-06-22) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=3.4 | 0=3.3 | 2=2.15
5. Algeria vs Austria (2026-06-27) [No bet]
    Venue: Kansas City
    Odds: 1=2.6 | 0=3.2 | 2=2.8
6. Jordan vs Argentina (2026-06-27) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=15.0 | 0=6.5 | 2=1.18

Match? (0 = back): 5

Selected: Algeria vs Austria
Venue: Kansas City
Odds: 1 (Home) 2.6 | 0 (Draw) 3.2 | 2 (Away) 2.8

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
2
Selected: 2 (Away Win) @ odds 2.8
Enter bet amount ($): 250

Potential win: $700.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $250.0 on 2 (Away Win) @ 2.8

--- Group J Matches ---
0. Back to group selection
1. Argentina vs Algeria (2026-06-16) [Bet: 1 (Home Win) $$500.00]
    Venue: Kansas City
    Odds: 1=1.25 | 0=5.5 | 2=12.0
2. Austria vs Jordan (2026-06-16) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=1.35 | 0=4.5 | 2=9.0
3. Argentina vs Austria (2026-06-22) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=1.4 | 0=4.5 | 2=8.0
4. Jordan vs Algeria (2026-06-22) [No bet]
    Venue: San Francisco Bay Area (Santa Clara)
    Odds: 1=3.4 | 0=3.3 | 2=2.15
5. Algeria vs Austria (2026-06-27) [Bet: 2 (Away Win) $$250.00]
    Venue: Kansas City
    Odds: 1=2.6 | 0=3.2 | 2=2.8
6. Jordan vs Argentina (2026-06-27) [No bet]
    Venue: Dallas (Arlington)
    Odds: 1=15.0 | 0=6.5 | 2=1.18

Match? (0 = back): 0
Choose by number (1-12) or letter (A-L)

[1/A] Group A              [2/B] Group B              [3/C] Group C
   Czech Republic             Bosnia & Herzegovina       Brazil
```

```
   South Korea                Canada                     Haiti
   Mexico                     Qatar                      Morocco
   South Africa               Switzerland                Scotland

[4/D] Group D              [5/E] Group E              [6/F] Group F
   Australia                   C u r a ao                  Japan
   Paraguay                   Ecuador                    Netherlands
   Turkey                     Germany                    Sweden
   USA                        Ivory Coast                Tunisia

[7/G] Group G              [8/H] Group H              [9/I] Group I
   Belgium                    Cape Verde                 France
   Egypt                      Saudi Arabia               Iraq
   Iran                       Spain                      Norway
   New Zealand                Uruguay                    Senegal

[10/J] Group J             [11/K] Group K             [12/L] Group L
   Algeria                    Colombia                   Croatia
   Argentina                  DR Congo                   England
   Austria                    Portugal                   Ghana
   Jordan                     Uzbekistan                 Panama

Group? (0 = back): 0

=====================================================
BEFORE GROUP STAGE
=====================================================
1) Show Standings (Pre-Tournament)
2) Show Matches
3) Place Bets
4) View / Modify Bets
5) Proceed to Group Stage
6) Exit
=====================================================
Choose: 4

=== Your Current Bets ===
0. Back
1. Argentina vs Algeria (2026-06-16) ? 1 (Home Win) @ 1.25 | $$500.00
2. Algeria vs Austria (2026-06-27) ? 2 (Away Win) @ 2.8 | $$250.00

Bet to modify? (0 = back): 0

=====================================================
BEFORE GROUP STAGE
=====================================================
1) Show Standings (Pre-Tournament)
2) Show Matches
3) Place Bets
4) View / Modify Bets
5) Proceed to Group Stage
6) Exit
=====================================================
Choose: 5

=====================================================
GROUP STAGE OVER
=====================================================
1) Group Standings
2) Bets Placed
3) Proceed to Round of 32
4) Exit
=====================================================
Choose: 2

=== Group Stage Bet Results ===
Total wagered: $750.00
Total won: $625.00
Net LOSS: $125.00
Correct: 1 | Incorrect: 1
Total bets: 2
Success rate: 50%

```

```
--- Detailed Results ---
Argentina vs Algeria ? You: 1 (Home Win) @ 1.25 | $$500.00 | ? WON | +$125.00
Algeria vs Austria ? You: 2 (Away Win) @ 2.8 | $$250.00 | ? LOST | -$250.00

=======================================================
GROUP STAGE OVER
=======================================================
1) Group Standings
2) Bets Placed
3) Proceed to Round of 32
4) Exit
=======================================================
Choose: 3

=======================================================
BEFORE ROUND OF 32
=======================================================
1) Show Matches
2) Place Bets
3) View / Modify Bets
4) Proceed
5) Exit
=======================================================
Choose: 4

=======================================================
ROUND OF 32 OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 3

=======================================================
BEFORE ROUND OF 16
=======================================================
1) Show Matches
2) Place Bets
3) View / Modify Bets
4) Proceed
5) Exit
=======================================================
Choose: 1

--- Matches ---
1. 2026-07-04: Mexico vs Canada [Philadelphia]
    Mexico: 4W-0D-0L, 12pts, GD:+5 (4 matches)
    Canada: 3W-1D-0L, 10pts, GD:+5 (4 matches)
2. 2026-07-04: Brazil vs USA [Houston]
    Brazil: 4W-0D-0L, 12pts, GD:+9 (4 matches)
    USA: 3W-1D-0L, 10pts, GD:+5 (4 matches)
3. 2026-07-05: Germany vs Netherlands [New York/New Jersey (East Rutherford)]
    Germany: 4W-0D-0L, 12pts, GD:+12 (4 matches)
    Netherlands: 4W-0D-0L, 12pts, GD:+7 (4 matches)
4. 2026-07-05: Belgium vs Spain [Mexico City]
    Belgium: 4W-0D-0L, 12pts, GD:+9 (4 matches)
    Spain: 3W-1D-0L, 10pts, GD:+9 (4 matches)
5. 2026-07-06: France vs Argentina [Dallas (Arlington)]
    France: 4W-0D-0L, 12pts, GD:+11 (4 matches)
    Argentina: 4W-0D-0L, 12pts, GD:+11 (4 matches)
6. 2026-07-06: Portugal vs England [Seattle]
    Portugal: 4W-0D-0L, 12pts, GD:+8 (4 matches)
    England: 4W-0D-0L, 12pts, GD:+7 (4 matches)
7. 2026-07-07: South Korea vs Iran [Atlanta]
    South Korea: 2W-1D-1L, 7pts, GD:+2 (4 matches)
    Iran: 2W-1D-1L, 7pts, GD:-1 (4 matches)
8. 2026-07-07: Ghana vs Ivory Coast [Vancouver]
    Ghana: 2W-1D-1L, 7pts, GD:+2 (4 matches)
    Ivory Coast: 2W-0D-2L, 6pts, GD:+1 (4 matches)

=======================================================
```

```
BEFORE ROUND OF 16
=======================================================
1) Show Matches
2) Place Bets
3) View / Modify Bets
4) Proceed
5) Exit
=======================================================
Choose: 2

--- Place Bets ---
0. Back
1. Mexico vs Canada (2026-07-04) [No bet]
2. Brazil vs USA (2026-07-04) [No bet]
3. Germany vs Netherlands (2026-07-05) [No bet]
4. Belgium vs Spain (2026-07-05) [No bet]
5. France vs Argentina (2026-07-06) [No bet]
6. Portugal vs England (2026-07-06) [No bet]
7. South Korea vs Iran (2026-07-07) [No bet]
8. Ghana vs Ivory Coast (2026-07-07) [No bet]

Match? (0 = back): 5

Selected: France vs Argentina
Venue: Dallas (Arlington) | Date: 2026-07-06
France: 4W-0D-0L, 12pts, GD:+11 (4 matches)
Argentina: 4W-0D-0L, 12pts, GD:+11 (4 matches)
Odds: 1 (Home) 1.85 | 0 (Draw) 3.4 | 2 (Away) 4.4

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
1
Selected: 1 (Home Win) @ odds 1.85
Enter bet amount ($): 1500

Potential win: $2775.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $1500.0 on 1 (Home Win) @ 1.85

--- Place Bets ---
0. Back
1. Mexico vs Canada (2026-07-04) [No bet]
2. Brazil vs USA (2026-07-04) [No bet]
3. Germany vs Netherlands (2026-07-05) [No bet]
4. Belgium vs Spain (2026-07-05) [No bet]
5. France vs Argentina (2026-07-06) [Bet: 1 (Home Win) $$1500.00]
6. Portugal vs England (2026-07-06) [No bet]
7. South Korea vs Iran (2026-07-07) [No bet]
8. Ghana vs Ivory Coast (2026-07-07) [No bet]

Match? (0 = back): 8

Selected: Ghana vs Ivory Coast
Venue: Vancouver | Date: 2026-07-07
Ghana: 2W-1D-1L, 7pts, GD:+2 (4 matches)
Ivory Coast: 2W-0D-2L, 6pts, GD:+1 (4 matches)
Odds: 1 (Home) 2.4 | 0 (Draw) 3.2 | 2 (Away) 3.0

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
3
Invalid input. No changes made.

--- Place Bets ---
0. Back
1. Mexico vs Canada (2026-07-04) [No bet]
2. Brazil vs USA (2026-07-04) [No bet]
3. Germany vs Netherlands (2026-07-05) [No bet]
4. Belgium vs Spain (2026-07-05) [No bet]
5. France vs Argentina (2026-07-06) [Bet: 1 (Home Win) $$1500.00]
6. Portugal vs England (2026-07-06) [No bet]
7. South Korea vs Iran (2026-07-07) [No bet]
8. Ghana vs Ivory Coast (2026-07-07) [No bet]
```

```

Match? (0 = back): 8

Selected: Ghana vs Ivory Coast
Venue: Vancouver | Date: 2026-07-07
Ghana: 2W-1D-1L, 7pts, GD:+2 (4 matches)
Ivory Coast: 2W-0D-2L, 6pts, GD:+1 (4 matches)
Odds: 1 (Home) 2.4 | 0 (Draw) 3.2 | 2 (Away) 3.0

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
2
Selected: 2 (Away Win) @ odds 3.0
Enter bet amount ($): 250

Potential win: $750.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $250.0 on 2 (Away Win) @ 3.0

--- Place Bets ---
0. Back
1. Mexico vs Canada (2026-07-04) [No bet]
2. Brazil vs USA (2026-07-04) [No bet]
3. Germany vs Netherlands (2026-07-05) [No bet]
4. Belgium vs Spain (2026-07-05) [No bet]
5. France vs Argentina (2026-07-06) [Bet: 1 (Home Win) $$1500.00]
6. Portugal vs England (2026-07-06) [No bet]
7. South Korea vs Iran (2026-07-07) [No bet]
8. Ghana vs Ivory Coast (2026-07-07) [Bet: 2 (Away Win) $$250.00]

Match? (0 = back): 0

=======================================================
BEFORE ROUND OF 16
=======================================================
1) Show Matches
2) Place Bets
3) View / Modify Bets
4) Proceed
5) Exit
=======================================================
Choose: 4

=======================================================
ROUND OF 16 OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 1

--- Round of 16 Results ---
2026-07-04: Mexico 2-1 Canada ? Mexico
    Mexico: 5W-0D-0L, 15pts, GD:+6 (5 matches)
    Canada: 3W-1D-1L, 10pts, GD:+4 (5 matches)
2026-07-04: Brazil 3-0 USA ? Brazil
    Brazil: 5W-0D-0L, 15pts, GD:+12 (5 matches)
    USA: 3W-1D-1L, 10pts, GD:+2 (5 matches)
2026-07-05: Germany 2-1 Netherlands ? Germany
    Germany: 5W-0D-0L, 15pts, GD:+13 (5 matches)
    Netherlands: 4W-0D-1L, 12pts, GD:+6 (5 matches)
2026-07-05: Belgium 1-2 Spain ? Spain
    Belgium: 4W-0D-1L, 12pts, GD:+8 (5 matches)
    Spain: 4W-1D-0L, 13pts, GD:+10 (5 matches)
2026-07-06: France 2-1 Argentina ? France
    France: 5W-0D-0L, 15pts, GD:+12 (5 matches)
    Argentina: 4W-0D-1L, 12pts, GD:+10 (5 matches)
2026-07-06: Portugal 0-1 England ? England
    Portugal: 4W-0D-1L, 12pts, GD:+7 (5 matches)
    England: 5W-0D-0L, 15pts, GD:+8 (5 matches)
```

```
2026-07-07: South Korea 2-0 Iran ? South Korea
    South Korea: 3W-1D-1L, 10pts, GD:+4 (5 matches)
    Iran: 2W-1D-2L, 7pts, GD:-3 (5 matches)
2026-07-07: Ghana 1-0 Ivory Coast ? Ghana
    Ghana: 3W-1D-1L, 10pts, GD:+3 (5 matches)
    Ivory Coast: 2W-0D-3L, 6pts, GD:+0 (5 matches)

=======================================================
ROUND OF 16 OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 2

=== Round of 16 Bet Results ===
Total wagered: $1750.00
Total won: $2775.00
Net PROFIT: $1025.00
Correct: 1 | Incorrect: 1
Total bets: 2
Success rate: 50%

--- Detailed Results ---
France vs Argentina ? You: 1 (Home Win) @ 1.85 | $$1500.00 | ? WON | +$1275.00
Ghana vs Ivory Coast ? You: 2 (Away Win) @ 3.0 | $$250.00 | ? LOST | -$250.00

=======================================================
ROUND OF 16 OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 3

=======================================================
BEFORE QUARTER-FINALS
=======================================================

1. Mexico vs Brazil
   Venue: Boston (Foxborough) | Date: 2026-07-09
   Mexico: 5W-0D-0L, 15pts, GD:+6 (5 matches)
   Brazil: 5W-0D-0L, 15pts, GD:+12 (5 matches)
   Odds: 1=3.2 | 0=3.3 | 2=2.25 [No bet]
   Enter b1 to bet on this match

2. Germany vs Spain
   Venue: Los Angeles (Inglewood) | Date: 2026-07-10
   Germany: 5W-0D-0L, 15pts, GD:+13 (5 matches)
   Spain: 4W-1D-0L, 13pts, GD:+10 (5 matches)
   Odds: 1=1.9 | 0=3.4 | 2=4.2 [No bet]
   Enter b2 to bet on this match

3. France vs England
   Venue: Miami (Miami Gardens) | Date: 2026-07-11
   France: 5W-0D-0L, 15pts, GD:+12 (5 matches)
   England: 5W-0D-0L, 15pts, GD:+8 (5 matches)
   Odds: 1=1.8 | 0=3.4 | 2=4.6 [No bet]
   Enter b3 to bet on this match

4. South Korea vs Ghana
   Venue: Kansas City | Date: 2026-07-11
   South Korea: 3W-1D-1L, 10pts, GD:+4 (5 matches)
   Ghana: 3W-1D-1L, 10pts, GD:+3 (5 matches)
   Odds: 1=1.7 | 0=3.6 | 2=5.0 [No bet]
   Enter b4 to bet on this match

0. Proceed
x. Exit
```

```
Choose: b3

Betting on: France vs England

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
1
Selected: 1 (Home Win) @ odds 1.8
Enter bet amount ($): 600

Potential win: $1080.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $600.0 on 1 (Home Win) @ 1.8

=======================================================
BEFORE QUARTER-FINALS
=======================================================

1. Mexico vs Brazil
   Venue: Boston (Foxborough) | Date: 2026-07-09
   Mexico: 5W-0D-0L, 15pts, GD:+6 (5 matches)
   Brazil: 5W-0D-0L, 15pts, GD:+12 (5 matches)
   Odds: 1=3.2 | 0=3.3 | 2=2.25 [No bet]
   Enter b1 to bet on this match

2. Germany vs Spain
   Venue: Los Angeles (Inglewood) | Date: 2026-07-10
   Germany: 5W-0D-0L, 15pts, GD:+13 (5 matches)
   Spain: 4W-1D-0L, 13pts, GD:+10 (5 matches)
   Odds: 1=1.9 | 0=3.4 | 2=4.2 [No bet]
   Enter b2 to bet on this match

3. France vs England
   Venue: Miami (Miami Gardens) | Date: 2026-07-11
   France: 5W-0D-0L, 15pts, GD:+12 (5 matches)
   England: 5W-0D-0L, 15pts, GD:+8 (5 matches)
   Odds: 1=1.8 | 0=3.4 | 2=4.6 [Bet: 1 (Home Win) $$600.00]
   Enter b3 to bet on this match

4. South Korea vs Ghana
   Venue: Kansas City | Date: 2026-07-11
   South Korea: 3W-1D-1L, 10pts, GD:+4 (5 matches)
   Ghana: 3W-1D-1L, 10pts, GD:+3 (5 matches)
   Odds: 1=1.7 | 0=3.6 | 2=5.0 [No bet]
   Enter b4 to bet on this match

0. Proceed
x. Exit
Choose: 0

=======================================================
QUARTER-FINALS OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 1

--- Quarter-final Results ---
2026-07-09: Mexico 2-3 Brazil ? Brazil
    Mexico: 5W-0D-1L, 15pts, GD:+5 (6 matches)
    Brazil: 6W-0D-0L, 18pts, GD:+13 (6 matches)
2026-07-10: Germany 1-2 Spain ? Spain
    Germany: 5W-0D-1L, 15pts, GD:+12 (6 matches)
    Spain: 5W-1D-0L, 16pts, GD:+11 (6 matches)
2026-07-11: France 1-0 England ? France
    France: 6W-0D-0L, 18pts, GD:+13 (6 matches)
    England: 5W-0D-1L, 15pts, GD:+7 (6 matches)
2026-07-11: South Korea 2-1 Ghana ? South Korea
    South Korea: 4W-1D-1L, 13pts, GD:+5 (6 matches)
```

```
    Ghana: 3W-1D-2L, 10pts, GD:+2 (6 matches)

=======================================================
QUARTER-FINALS OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 2

=== Quarter-final Bet Results ===
Total wagered: $600.00
Total won: $1080.00
Net PROFIT: $480.00
Correct: 1 | Incorrect: 0
Total bets: 1
Success rate: 100%

--- Detailed Results ---
France vs England ? You: 1 (Home Win) @ 1.8 | $$600.00 | ? WON | +$480.00

=======================================================
QUARTER-FINALS OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 3

=======================================================
BEFORE SEMI-FINALS
=======================================================

1. Brazil vs Spain
   Venue: Dallas (Arlington) | Date: 2026-07-14
   Brazil: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   Spain: 5W-1D-0L, 16pts, GD:+11 (6 matches)
   Odds: 1=1.75 | 0=3.5 | 2=4.8 [No bet]
   Enter b1 to bet on this match

2. France vs South Korea
   Venue: Atlanta | Date: 2026-07-15
   France: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   South Korea: 4W-1D-1L, 13pts, GD:+5 (6 matches)
   Odds: 1=1.3 | 0=5.0 | 2=10.0 [No bet]
   Enter b2 to bet on this match

0. Proceed
x. Exit
Choose: 2
Invalid option.

=======================================================
BEFORE SEMI-FINALS
=======================================================

1. Brazil vs Spain
   Venue: Dallas (Arlington) | Date: 2026-07-14
   Brazil: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   Spain: 5W-1D-0L, 16pts, GD:+11 (6 matches)
   Odds: 1=1.75 | 0=3.5 | 2=4.8 [No bet]
   Enter b1 to bet on this match

2. France vs South Korea
   Venue: Atlanta | Date: 2026-07-15
   France: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   South Korea: 4W-1D-1L, 13pts, GD:+5 (6 matches)
   Odds: 1=1.3 | 0=5.0 | 2=10.0 [No bet]
   Enter b2 to bet on this match
```

```

0. Proceed
x. Exit
Choose: b2

Betting on: France vs South Korea

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
1
Selected: 1 (Home Win) @ odds 1.3
Enter bet amount ($): 1000

Potential win: $1300.00
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $1000.0 on 1 (Home Win) @ 1.3

=======================================================
BEFORE SEMI-FINALS
=======================================================

1. Brazil vs Spain
   Venue: Dallas (Arlington) | Date: 2026-07-14
   Brazil: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   Spain: 5W-1D-0L, 16pts, GD:+11 (6 matches)
   Odds: 1=1.75 | 0=3.5 | 2=4.8 [No bet]
   Enter b1 to bet on this match

2. France vs South Korea
   Venue: Atlanta | Date: 2026-07-15
   France: 6W-0D-0L, 18pts, GD:+13 (6 matches)
   South Korea: 4W-1D-1L, 13pts, GD:+5 (6 matches)
   Odds: 1=1.3 | 0=5.0 | 2=10.0 [Bet: 1 (Home Win) $$1000.00]
   Enter b2 to bet on this match

0. Proceed
x. Exit
Choose: 0

=======================================================
SEMI-FINALS OVER
=======================================================
1) Match Results
2) Bets Placed
3) Proceed
4) Exit
=======================================================
Choose: 3

=======================================================
BEFORE FINAL
=======================================================

1. Brazil vs France
   Venue: New York/New Jersey (East Rutherford) | Date: 2026-07-19
   Brazil: 7W-0D-0L, 21pts, GD:+14 (7 matches)
   France: 7W-0D-0L, 21pts, GD:+16 (7 matches)
   Odds: 1=2.1 | 0=3.3 | 2=3.5 [No bet]
   Enter b1 to bet on this match

0. Proceed
x. Exit
Choose: b1

Betting on: Brazil vs France

Enter prediction: 1=Home Win, 2=Away Win, 0=Draw, x=Remove bet
1
Selected: 1 (Home Win) @ odds 2.1
Enter bet amount ($): 650

Potential win: $1365.00
```

```
1) Confirm
2) Edit
Choose: 1
Bet confirmed: $650.0 on 1 (Home Win) @ 2.1

=====================================================
BEFORE FINAL
=====================================================

1. Brazil vs France
   Venue: New York/New Jersey (East Rutherford) | Date: 2026-07-19
   Brazil: 7W-0D-0L, 21pts, GD:+14 (7 matches)
   France: 7W-0D-0L, 21pts, GD:+16 (7 matches)
   Odds: 1=2.1 | 0=3.3 | 2=3.5 [Bet: 1 (Home Win) $$650.00]
   Enter b1 to bet on this match

0. Proceed
x. Exit
Choose: 0


=====================================================
TOURNAMENT OVER
=====================================================

CHAMPION: France
Final: Brazil 2-3 France

Total wagered: $4750.00
Total won: $5780.00
Net PROFIT: $1030.00

Thank you for playing!
```

Listing 2: Full Program Output - Final Version

GitHub Repository Link:
`https://github.com/RohitSonejee0112/AISE26_Assignment1`

Declaration Confirmation:
I have read and understood the assignment guidelines. All use of generative AI has been transparently declared above.

Rohit Sonejee
Matrikelnummer: 4756184